ENCS304 – Computer Networks

Assignment 1: Implementation and Analysis of Basic Network Topologies

Student Details

Name: Shikhar Bajpai
Roll No.: 2301010188
Programme: B.Tech Computer Science & Engineering
Section: Section 3
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1. Objective

The objective of this experiment is to design and simulate different basic LAN topologies using Cisco Packet Tracer and analyze their performance, communication behavior, and fault tolerance.

The experiment focuses on comparing switch-based, hub-based, and ring-like network structures using IPv4 addressing and ICMP ping testing.

2. Software Used

- Cisco Packet Tracer (Version 8.0 or above)

3. Task 1 – Star Topology Using Switch

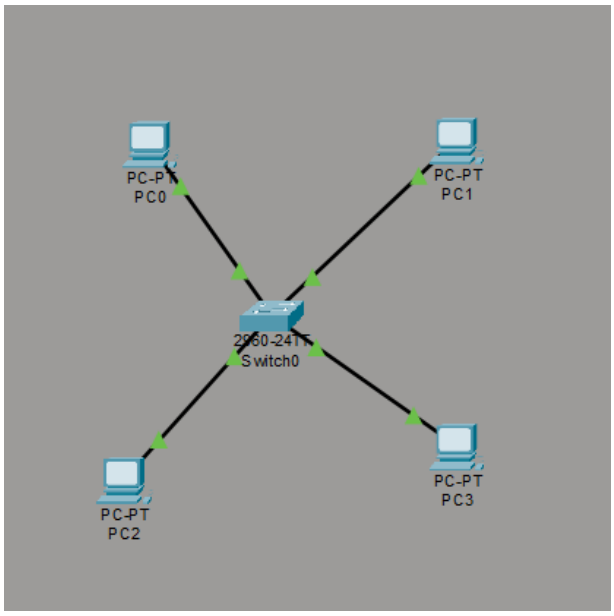
3.1 Topology Description

In this setup, one switch (2960-24TT) was connected to four PCs using straight-through cables.
All devices were configured in the same subnet: **10.10.10.0/24**.

3.2 IP Addressing Table

Device	IP Address	Subnet Mask
PC0	10.10.10.1	255.255.255.0
PC1	10.10.10.2	255.255.255.0
PC2	10.10.10.3	255.255.255.0
PC3	10.10.10.4	255.255.255.0

3.3 Screenshot



3.4 Ping Output

```
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.10.3

Pinging 10.10.10.3 with 32 bytes of data:

Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time=1ms TTL=128
Reply from 10.10.10.3: bytes=32 time<1ms TTL=128
Reply from 10.10.10.3: bytes=32 time=7ms TTL=128

Ping statistics for 10.10.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>ping 10.10.10.4

Pinging 10.10.10.4 with 32 bytes of data:

Reply from 10.10.10.4: bytes=32 time<1ms TTL=128
Reply from 10.10.10.4: bytes=32 time<1ms TTL=128
Reply from 10.10.10.4: bytes=32 time<1ms TTL=128
Reply from 10.10.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3.5 Observation

All devices communicated successfully with zero packet loss.
The switch initially broadcasted frames for unknown MAC addresses but quickly learned them and updated its MAC address table.
After learning, frames were forwarded only to the destination port, improving network efficiency and reducing unnecessary traffic.

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4. Task 2 – Hub-Based Topology

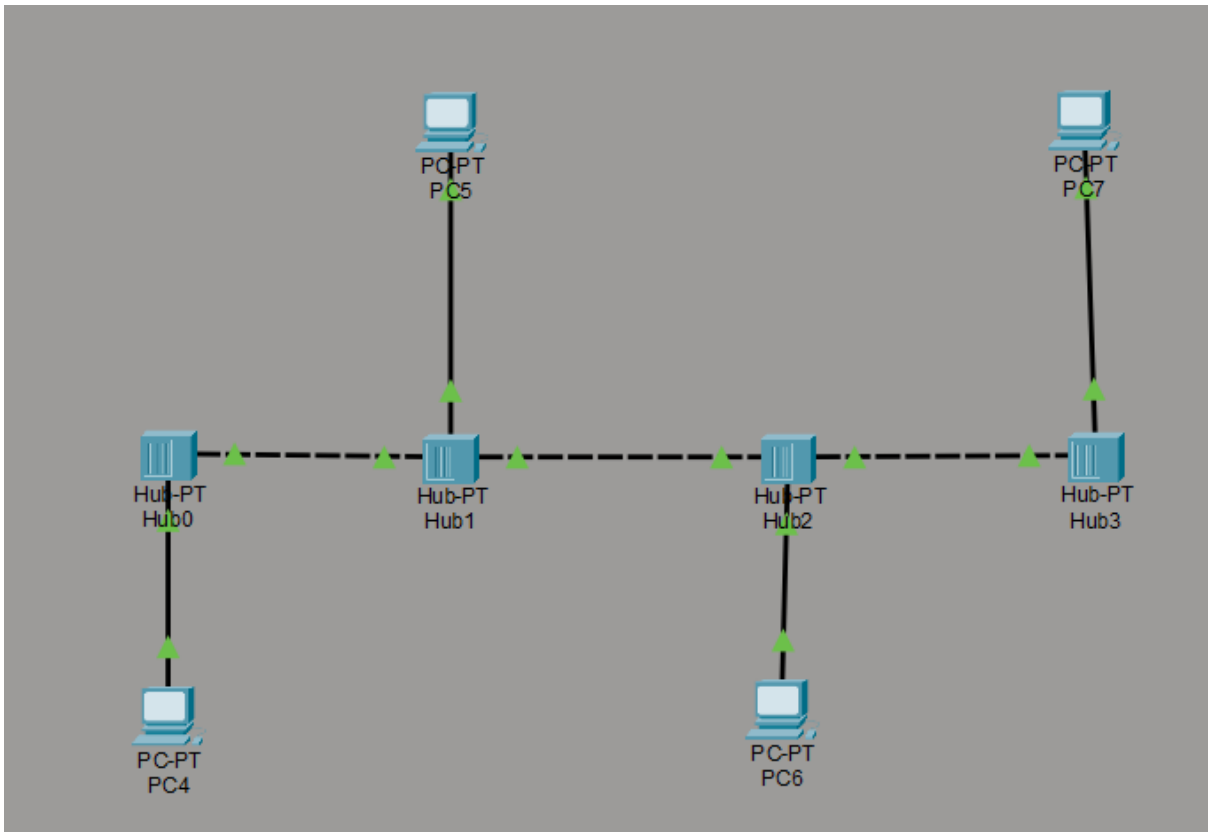
4.1 Topology Description

In this setup, the switch was replaced by a hub.
Four PCs were connected to the hub using straight-through cables.
All devices were configured in the subnet **10.10.10.11 – 10.10.10.14 /24**.

4.2 IP Addressing Table

Device	IP Address	Subnet Mask
PC0	10.10.10.11	255.255.255.0
PC1	10.10.10.12	255.255.255.0
PC2	10.10.10.13	255.255.255.0
PC3	10.10.10.14	255.255.255.0

4.3 Screenshot



4.4 Ping Output

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.12

Pinging 10.10.10.12 with 32 bytes of data:

Reply from 10.10.10.12: bytes=32 time=2ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time=1ms TTL=128

Ping statistics for 10.10.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

4.5 Observation

All devices were able to communicate successfully.
However, in simulation mode it was observed that the hub forwards incoming frames to every connected device without filtering.
This creates unnecessary traffic and may cause collisions in larger networks.
Compared to the switch-based topology, the hub-based network is less efficient and less scalable.

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5. Task 3 – Ring-like Topology (Switch Loop with STP)

5.1 Topology Description

Three switches were connected in a loop configuration.
Each switch had one or more PCs connected to it.
All devices were configured in the subnet **10.10.10.21 – 10.10.10.23 /24**.

Spanning Tree Protocol (STP) was automatically enabled on the switches.

5.2 Screenshot

(Insert Ring Topology Screenshot Here)

5.3 Ping Output

```
C:\>ping 10.10.10.25

Pinging 10.10.10.25 with 32 bytes of data:

Reply from 10.10.10.25: bytes=32 time=16ms TTL=128
Reply from 10.10.10.25: bytes=32 time<1ms TTL=128
Reply from 10.10.10.25: bytes=32 time<1ms TTL=128
Reply from 10.10.10.25: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.25:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 16ms, Average = 4ms
```

5.4 Observation

Communication between devices connected to different switches was successful.
STP detected the loop and automatically blocked one redundant link to prevent broadcast storms.
The network remained stable and functional with zero packet loss.

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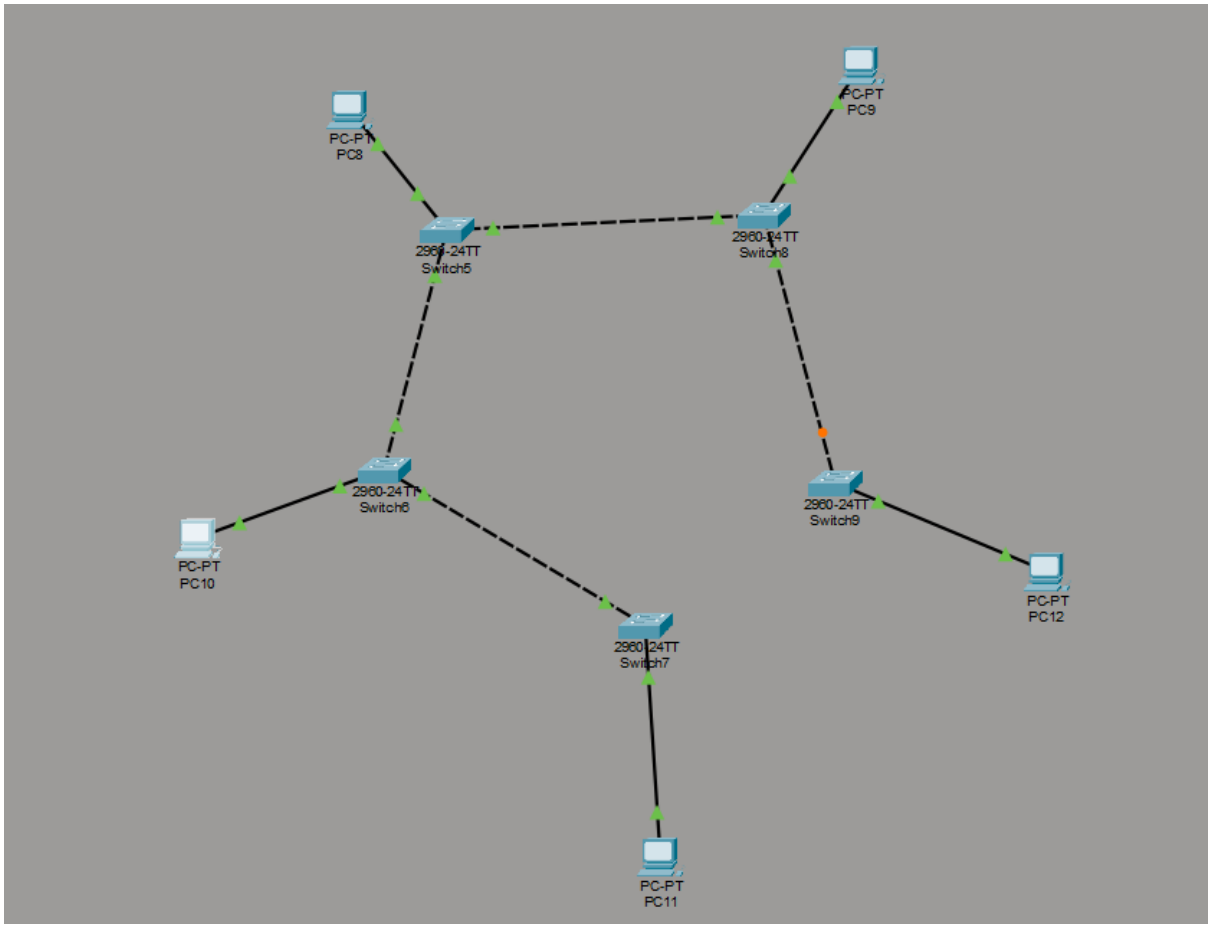
6. Task 4 – Link Failure Test

6.1 Procedure

One inter-switch link was manually disconnected to simulate a network failure.
After a few seconds, STP recalculated the topology and activated the previously blocked link.

6.2 Screenshot



6.3 Ping Output

```
C:\>ping 10.10.10.24

Pinging 10.10.10.24 with 32 bytes of data:

Reply from 10.10.10.24: bytes=32 time<1ms TTL=128
Reply from 10.10.10.24: bytes=32 time<1ms TTL=128
Reply from 10.10.10.24: bytes=32 time<1ms TTL=128
Reply from 10.10.10.24: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.24:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

6.4 Observation

Even after disconnecting one link, communication was restored through an alternate path.
STP ensured network stability by reconfiguring the topology automatically.
This demonstrates that ring topology with STP provides better fault tolerance compared to a simple star topology.

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7. Comparison of Topologies

Feature	Switch (Star)	Hub	Ring with STP
Efficiency	High	Low	High
Broadcast Traffic	Limited	High	Controlled
Fault Tolerance	Low	Low	High
Scalability	Good	Poor	Good

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8. Conclusion

In this experiment, different LAN topologies were implemented and analyzed using Cisco Packet Tracer.

The switch-based star topology provided better performance due to MAC address learning and selective forwarding.
The hub-based topology showed increased broadcast traffic and lower efficiency.
The ring-like topology with STP demonstrated fault tolerance by maintaining communication even after link failure.

This experiment helped in understanding the practical differences between network topologies and their impact on performance and reliability in real-world LAN environments.